

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
1st Semester of M. Pharm
(Pharmaceutical Technology and Pharmaceutics (Drug Regulatory Affairs))
University Theory Examination March-April 2018
PHCC002- BIOPHARMACEUTICS AND PHARMACOKINETICS

Date: 27.03.2018 , Tuesday Time: 10:00 a.m to 1:00 p.m Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1** **A)** Define drug absorption. Discuss the factors affecting absorption process of drug molecules . **05**
- B)** Explain the oral PBPK model with figure for ADME process in detail. **05**
- Q.2** **A)** Enlists various method of determining the absorption of drug. Explain Wagner-Nelson method for estimation of absorption rate constant. **05**
- B)** What is the importance of measuring plasma drug concentration? Describe different types of Pharmacokinetics models. **05**

OR

- Q.2** **A)** Discuss factors affecting gastric emptying. Describe conditions in which rapid and delayed gastric emptying is desired? **05**
- B)** Explain the drug distribution with role of various active transporters. **05**
- Q.3** **Answer the following (Any three)** **15**
- (A)** Write a note on nonlinear pharmacokinetics.
- (B)** Explain in detail the factors affecting biotransformation.
- (C)** Discuss in detail therapeutic drug monitoring.
- (D)** What is Biowaiver? What is the criteria for this?

SECTION-II

- Q.4** **A)** Discuss the importance of BCS in dosage form development. **05**
- B)** Write a note on methods of establishing IVIVC. **05**

- Q.5** **A)** Define apparent volume of distribution, biological half life and total body clearance. Describe methods to determine V_d of the new drug molecule. **10**

OR

- Q.5** **A)** What is multicompartment model? Enlist such multicompartment models and write a note on two compartment model. **10**

- Q.6** **Answer the following (Any three)** **15**
- (A)** Explain the importance of bioequivalence study .
- (B)** Discuss pharmaceutical consideration for CACO2.
- (C)** Derive equation for one compartment open model- I.V. bolus.
- (D)** What are limitations and applications of method of residuals?